

MagiQal Encryption

Aiming at weeding out the weakest link in the chain of encryption, a company called MagiQ Technologies Inc., claims to have developed an unbreakable and unbreachable encryption system, named Navajo. The Navajo black box system costs about \$100,000 (Rs 4,55,100), and can span 70 miles, after which additional relay boxes are needed. The system comprises 19-inch boxes which communicate with each other over optical lines.

Navajo is secure because it uses Quantum encryption. The keys are transmitted using photons, which are discrete particles of energy, and they change 10 times a second. This means if a person gets hold of the key, it will be rendered useless almost immediately. Also, based on Heisenberg's Uncertainty Principle, the extremely sensitive photons change their behaviour if somebody tries to spy on the communication during the transfer.

Up until now, people have only been using bigger and bigger encryption keys generated by more and more complex algorithms, which means that breaking them was just a question of 'when'. Quantum cryptography changes all that.

snapshot

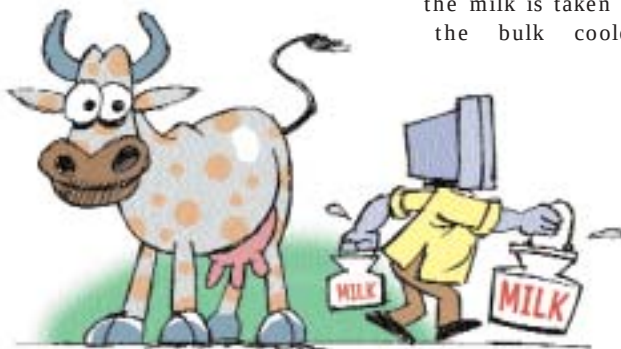
Largest prime number with
63,20,430 decimals calculated using
2,00,000 computers

Source: Great Internet Mersenne Prime Search

Analysing milk, PC-style!

In Baramati, Maharashtra, the life of dairy-farmers has changed in recent times. Earlier, cow owners had to milk cows and rush to the local market to sell the milk before it got sour. At times, almost 50

Cows are milked by a machine, and the milk is taken to the bulk coolers. The fat content of the milk is checked by a computer, and the price accordingly determined. The dairy farmer is paid immediately, and the milk is taken to the bulk cooler.



per cent would be left unsold. However, with the efforts of Dynamix Technologies, which sells milk to Nestlé and Britannia, things have changed. It has set up computerised and refrigerated milk collection centres, locally known as bulk coolers.

Later, it is transported to the Dynamix factory.

Computers are also used to maintain a database of the complete history of health, nutrition, disease, pregnancy etc of a cow. This helps dairy farmers develop better breeds that produce more milk.

tomorrow's technology

Chip off its size!

Two recent technological advances will allow shrinking of chips to continue for some time. NEC announced it has developed the world's smallest transistor that is 1/18th the size of existing transistors. Using the new technology, a chip measuring 1 sq. cm will be able to hold 40 billion transistors—about 150 times more than the current density. Such transistors can be used to build a supercomputer that churns out 600 billion operations a second, while

being the size of an ordinary desktop.

The company expects to start production by 2020.

In a separate development, IBM announced it uses molecular self assembly to build smaller chips. Organic polymer molecules assemble themselves into hexagonal patterns that are denser and more precise than those obtained using lithography. The process can be implemented using existing manufacturing tools, making it inexpensive.

quoteworthy

Everyone's going to be talking about the digital divide, and how massive it is. Is the digital divide as big as we think? The answer is, 'No!'

Michael Minges, co-author of the annual study by the International Telecommunication Union

US society is not being re-skilled and re-tooled to stay on the top of the emerging environment

Wipro Chairman, Azim Premji, on why IT firms find India attractive

They are winning by a considerable margin. Very few have been identified, prosecuted or punished

David Finn, Director of digital integrity, Microsoft, commenting about pirates in South-east Asia who seem to be winning the war on piracy

It's a dirty secret. Floating-point arithmetic is wrong. It only takes two operations to see that computers make mistakes with fractions.

John Gustafson, an investigator with Sun Microsystems, Santa Clara, California, on bad mathematical calculations being the reason behind system crashes. He reckons Microsoft need not always be the culprit